

Microeconomics

Lecture 17: Rent-Seeking, the Lerner Index and Price Discrimination

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What We Know

In the previous lecture we built the basic monopoly model:

- The monopolist is a price-maker and faces the downward-sloping market demand
- $MR < P$ always: to sell more, it must lower the price *on all units*
- Optimum: $MR = MC$, which gives Q_m^* and $P_m^* = P(Q_m^*) > MC(Q_m^*)$
- Compared with PC: lower Q , higher P , $\pi > 0$ even in the LR
- Monopoly creates $DWL = \frac{1}{2}(Q_{pc} - Q_m^*)(P_m^* - MC(Q_m^*))$

Today: three questions about market power

1. Is the monopolist's *rent rectangle* really just a transfer, or can it become a social cost? $\Rightarrow$ **rent-seeking**
2. How do we *measure* market power? $\Rightarrow$ **Lerner index**
3. Can the monopolist capture *even more* consumer surplus? $\Rightarrow$ **price discrimination**

Learning Objectives

1. **Distinguish** monopoly as a transfer from monopoly with rent-seeking
2. **Analyze** the Tullock model and understand why the rent can be *entirely dissipated*
3. **Define** the Lerner index and **derive** the relation $L = 1/|\varepsilon_D|$
4. **Distinguish** the three forms of discrimination (Pigou: first, second, third degree) and the necessary conditions
5. **Analyze** perfect discrimination: capturing all of the CS, eliminating the DWL
6. **Compute** the optimum under third-degree discrimination: $MR_1 = MR_2 = MC$
7. **Recognize** the forms of second-degree discrimination: two-part tariffs, bundling, versioning
8. **Construct** a two-part tariff with *homogeneous* and *heterogeneous* consumers
9. **Compare** welfare under the different pricing regimes (summary table)
10. **Solve** numerical exercises on first- and third-degree discrimination

Recap: Is the Monopoly Rent a *Transfer*?

In the standard model of L16, the rectangle $\pi_m = (P_m^* - AC(Q_m^*)) \cdot Q_m^*$ is a **transfer** from consumers to the monopolist: it changes who owns the surplus, but does not destroy it.

Apparent implication: the social cost of monopoly is *only* the *DWL* triangle (the area of forgone trades).

This view is incomplete

The rent π_m is a **prize** that someone is willing to compete for. If firms invest *real* resources (lobbying, litigation, defensive advertising, regulatory barriers) to obtain or maintain the monopoly position, those resources are **diverted from productive uses**: the rectangle π_m can turn (wholly or partly) into a social cost.

Seminal references: Tullock (*Western Economic Journal*, 1967) and Krueger (*AER*, 1974). Krueger estimated that, in some developing economies, the rents associated with import licenses exceeded 7% of GDP.

What Is Rent-Seeking?

Rent-seeking

The investment of **socially unproductive** resources to obtain, maintain, or extend a position of advantage (monopoly, favorable regulation, exclusive access to a resource) that generates economic rents.

Concrete examples:

- **Lobbying:** political influence spending to obtain licenses, extended patents, protective tariffs, subsidies
- **Strategic litigation:** lawsuits to exclude competitors (*patent thickets*, abuse of dominant position)
- **Defensive advertising:** spending that does not shift overall demand but protects market shares
- **Exclusive contracts:** payments to suppliers/distributors to block competitors' entry
- **Regulatory capture:** investments to steer sector regulation

Crucial distinction

Investing in *R&D* to reduce *MC* is *productive* (it shifts the technological frontier). Investing in lobbying to obtain an extended patent is *unproductive* (it reduces competition). The difference is not the cost to the firm, but the effect on aggregate welfare.

The Tullock Model (1967)

Setup. n identical firms compete for a rent V (expected value of the monopoly profit). Each invests $x_i \geq 0$ in rent-seeking activities. The probability of obtaining the rent is proportional to spending:

$$p_i(x_1, \dots, x_n) = \frac{x_i}{\sum_{j=1}^n x_j}$$

Expected profit of firm i :

$$\pi_i = \frac{x_i}{\sum_j x_j} \cdot V - x_i$$

First-order condition (symmetric equilibrium, $x_i = x$ for all i):

$$\frac{\partial \pi_i}{\partial x_i} = \frac{\sum_{j \neq i} x_j}{(\sum_j x_j)^2} \cdot V - 1 = 0$$

Imposing symmetry, $\sum_j x_j = nx$ and $\sum_{j \neq i} x_j = (n-1)x$:

$$\frac{(n-1)x}{(nx)^2} \cdot V = 1 \implies \boxed{x^* = \frac{n-1}{n^2} V}$$

Tullock: How Much of the Rent Is Dissipated?

Aggregate spending in equilibrium:

$$X^* = n \cdot x^* = \frac{n-1}{n} V$$

Fraction of the rent dissipated:

n	x^*/V	X^*/V	Dissipation
2	0.25	0.50	50%
3	0.22	0.67	66.7%
5	0.16	0.80	80%
10	0.09	0.90	90%
$\rightarrow \infty$	$\rightarrow 0$	$\rightarrow 1$	$\rightarrow 100\%$

Key result

With large n the rent is **almost entirely dissipated**: the rectangle π_m turns into a real cost for society. Tullock and Krueger generalize: in the competitive limit, the *entire* monopoly rent is an additional cost.

Caveat. The result depends on the *contest success function* and on symmetry. With risk aversion or asymmetries, dissipation is generally *less* than V .

Welfare Implications: the Effective DWL

Without rent-seeking (L16 model):

$$DWL_{\text{base}} = \frac{1}{2}(Q_{pc} - Q_m^*)(P_m^* - MC) \quad [\text{Harberger triangle}]$$

With rent-seeking that dissipates a fraction $\delta \in [0, 1]$ of the rent:

$$DWL_{\text{eff}} = \underbrace{\frac{1}{2}(Q_{pc} - Q_m^*)(P_m^* - MC)}_{\text{triangle}} + \underbrace{\delta \cdot \pi_m}_{\text{dissipated rent}}$$

Numerical example (link to L16, $P = 70 - Q$, $C = Q^2 + 10Q + 50$): $DWL_{\text{base}} = 37.5$, $\pi_m = 400$. If $\delta = 0.9$ (10 competing firms):

$$DWL_{\text{eff}} = 37.5 + 0.9 \cdot 400 = 397.5$$

The effective social cost is more than **10 times** the standard triangle.

Policy lesson

The Harberger triangle severely underestimates the social cost of monopoly when there is competition for the rent. Antitrust and transparency policies (lobbying registers, patent scrutiny) are valuable not only because of the triangle, but above all because they prevent rent dissipation.

Rent-Seeking: Empirical Evidence and Policies

Size of the phenomenon (indicative estimates):

- Annual US federal lobbying spending: $\sim$ \$4 bn/year (*OpenSecrets*, 2024); EU lobbying register $\sim$ €1.8 bn/year
- Estimates of aggregate rent-seeking: 3–10% of GDP in advanced economies (survey in Murphy-Shleifer-Vishny, *AER* 1993)
- Krueger (1974): 7–15% of GDP in India and Turkey in the 1960s–1970s

Types of policy against rent-seeking:

- Transparency (lobbyist registers, conflict-of-interest disclosures)
- Limits on patent duration and review of dominant positions (EU DG Comp, US FTC)
- Reduction of regulatory barriers to entry (*red tape*, administrative simplification)
- Competition for the market (competitive auctions for concessions: spectrum, highways, lotteries)

Case study: 3G/4G spectrum auctions

Auctions for mobile telephony licenses (UMTS in the EU, 2000, $\sim$ €100 bn) transferred much of the monopoly rent to the public sector, reducing dissipation. By replacing lobbying with a transparent competitive mechanism, the State captures the rent instead of seeing it dissipated.

Measuring Market Power: the Lerner Index

Lerner index (1934)

The **Lerner index** measures the gap between price and marginal cost as a fraction of the price:

$$L = \frac{P - MC}{P} \in [0, 1]$$

Interpretation:

- $L = 0$: no market power (perfect competition: $P = MC$)
- $L = 1$: maximum market power ($MC = 0$, e.g. digital information)
- $L = 0.5$: the monopolist charges a markup equal to 50% of the final price

Relation with elasticity (derived on the next slide)

At the monopolist's optimum, $L = 1/|\varepsilon_D|$: *the more inelastic demand is, the higher the markup*. The monopolist exploits the inelasticity of demand to earn a margin above marginal cost.

Derivation of $L = 1/|\varepsilon_D|$

At the monopolist's optimum $MR = MC$. Substituting the formula $MR = P(1 - 1/|\varepsilon_D|)$ derived in L16:

$$P \left(1 - \frac{1}{|\varepsilon_D|} \right) = MC$$
$$P - \frac{P}{|\varepsilon_D|} = MC \implies P - MC = \frac{P}{|\varepsilon_D|}$$

$$L = \frac{P - MC}{P} = \frac{1}{|\varepsilon_D|}$$

Consequences

- High $|\varepsilon_D|$ (elastic demand) $\Rightarrow$ low L : the monopolist cannot charge a high markup without losing too many customers
- Low $|\varepsilon_D|$ (inelastic demand) $\Rightarrow$ high L : high markup
- Perfect-competition limit: $|\varepsilon_D| \rightarrow \infty$, so $L \rightarrow 0$ and $P \rightarrow MC$
- The monopolist never produces where $|\varepsilon_D| < 1$ (it would require $L > 1$, which is impossible)

Lerner Index: Check and Applications

Check on the L16 example ($P = 70 - Q$, $C(Q) = Q^2 + 10Q + 50$, $Q_m^* = 15$, $P_m^* = 55$):

$$MC(15) = 2(15) + 10 = 40, \quad L = \frac{55 - 40}{55} = \frac{15}{55} \approx 0.273$$

The monopolist charges a markup of 27.3%. Check with $|\varepsilon_D|$ at $Q = 15$, $P = 55$:

$$\varepsilon_D = \frac{dQ}{dP} \cdot \frac{P}{Q} = -1 \cdot \frac{55}{15} = -\frac{11}{3} \Rightarrow \frac{1}{|\varepsilon_D|} = \frac{3}{11} \approx 0.273 \checkmark$$

Sector applications (indicative values):

Sector	Estimated $ \varepsilon_D $	Lerner index L
Patented drugs	1.1–1.5	0.67–0.91
Cloud services (AWS, Azure)	1.5–2.0	0.50–0.67
Supermarkets (grocery)	3–5	0.20–0.33

Regulatory use

Antitrust authorities (FTC, AGCM, DG Comp) use Lerner-type indices to assess whether a firm has abusive market power. A high L is a *signal* (not proof) of a dominant position.

What Is Price Discrimination?

Price discrimination

Price discrimination occurs when a firm sells the same unit (or similar units) at different prices to different consumers (or for different quantities), *not justified by cost differences*.

Why does the monopolist want to discriminate? Standard (single-price) monopoly leaves $CS > 0$: consumers with high WTP pay the same price as those with low WTP. The monopolist would like to capture that surplus too.

Necessary conditions:

1. **Market power** (price-maker): under PC no discrimination is possible
2. **Information** about consumers' types/WTP (observable or inferable)
3. **No arbitrage** between groups (no resale from low payers to high payers)

Three degrees of discrimination (Pigou, 1920):

- **First degree** (perfect): price = each consumer's unit WTP
- **Second degree**: nonlinear prices (quantity discounts, two-part tariffs, bundling)
- **Third degree**: different prices for identifiable groups (students, business, countries, ...)

First-Degree (Perfect) Discrimination

First-degree discrimination

The monopolist knows each consumer's WTP for each unit and charges exactly that. Each consumer pays the **maximum price** he or she is willing to pay for each unit.

Effects:

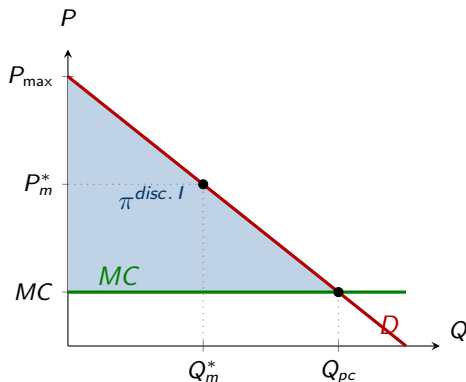
- The monopolist captures **all** of the consumer surplus: $CS = 0$
- The quantity produced is Q_{pc} (the same as under perfect competition): **no DWL**
- The monopolist's profit is maximal (upper bound of all discrimination strategies)
- Equivalently: the monopolist produces up to $P(Q) = MC(Q)$, because every extra unit with $WTP > MC$ is traded

Is it efficient? Yes in the Pareto sense (no forgone trades), but extremely **inequitable**: consumers get no benefit from trade, all the surplus goes to the monopolist.

Examples (approximate)

- **Individual bargaining**: a merchant negotiating the price with each customer
- **First-price auctions**: the winner pays his or her own bid, which approaches the reservation value when competition among bidders is intense
- **Personalized online prices**: platforms showing different prices based on the user profile

First-Degree Discrimination: Graph



Logic of the graph ($P = 100 - Q$, $MC = 20$):

- The monopolist sells each unit at the price on the demand curve
- It stops at $Q_{pc} = 80$, where $P = MC$
- The area $\pi^{disc. I}$ (triangle $D-MC$) is *all* profit

Comparison with a single price:

- Uniform mon.: $\pi_m = 1,600$, $CS = 800$, $DWL = 800$
- First-degree discr.: $\pi = 3,200$, $CS = 0$, $DWL = 0$

Interpretation. The monopolist captures both the CS triangle of the uniform model and the DWL triangle. No forgone trades.

Third-Degree Discrimination: Logic

Third-degree discrimination

The monopolist divides consumers into **identifiable groups** (by age, occupation, nationality, purchase channel, time of purchase) and sets a different price for each group.

Real-world examples:

- Cinema/museum tickets: reduced price for students and seniors
- Airfares: business class vs economy
- Software: academic vs commercial license
- Drugs: different prices in developed and developing countries

Optimum of the discriminating monopolist

With n separate markets and a common marginal cost MC :

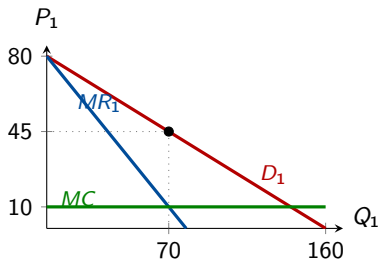
$$MR_1(Q_1) = MR_2(Q_2) = \dots = MR_n(Q_n) = MC$$

The monopolist sets a lower price in the market with more elastic demand and a higher price in the less elastic one. From the Lerner formula: $L_i = 1/|\varepsilon_{D,i}|$.

Intuition. In the less elastic market, cutting P is costly in revenue lost on units already sold: it pays to keep the price high.

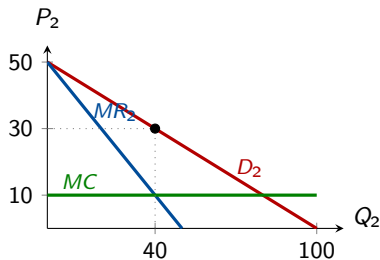
Third-Degree Discrimination: Graph

Market 1 (less elastic)



$$P_1 = 80 - Q_1/2, MR_1 = 80 - Q_1$$
$$Q_1^* = 70, P_1^* = 45, L_1 = 35/45 \approx 0.78$$

Market 2 (more elastic)



$$P_2 = 50 - Q_2/2, MR_2 = 50 - Q_2$$
$$Q_2^* = 40, P_2^* = 30, L_2 = 20/30 \approx 0.67$$

Third-Degree Discrimination: Numerical Example

Let us go back to the setup of the previous slide: $MC = 10$, two markets with demands

$$P_1 = 80 - Q_1/2 \Rightarrow MR_1 = 80 - Q_1, \quad P_2 = 50 - Q_2/2 \Rightarrow MR_2 = 50 - Q_2$$

Separate optimum. $MR_1 = MC$:

$$80 - Q_1 = 10 \Rightarrow Q_1^* = 70, \quad P_1^* = 80 - 35 = 45$$

$MR_2 = MC$:

$$50 - Q_2 = 10 \Rightarrow Q_2^* = 40, \quad P_2^* = 50 - 20 = 30$$

Consistency with Lerner. $L_1 = 35/45 \approx 0.78$, $L_2 = 20/30 \approx 0.67$. Computing elasticities:
 $|\varepsilon_1| = 2 \cdot 45/70 \approx 1.29 \rightarrow 1/|\varepsilon_1| = 0.78 \checkmark$; $|\varepsilon_2| = 2 \cdot 30/40 = 1.5 \rightarrow 1/|\varepsilon_2| = 0.67 \checkmark$.

Total profit (with fixed costs $FC = 200$):

$$\pi_1 = 35 \cdot 70 = 2,450, \quad \pi_2 = 20 \cdot 40 = 800, \quad \pi_{\text{tot}} = 2,450 + 800 - 200 = 3,050$$

Interpretation

Without discrimination, the monopolist would charge a single price to all consumers, giving up the possibility of extracting more surplus from the group with inelastic demand. Discrimination **always increases profit** (weakly) relative to a single price.

Second-Degree Discrimination (Nonlinear Prices)

Second-degree discrimination

The price per unit depends on the **quantity purchased** by the same consumer. It does not require identifying types in advance: consumers self-select by choosing from the “menu” offered.

Main forms:

- **Quantity discounts:** the first k units cost p_1 , subsequent ones $p_2 < p_1$ (e.g. phone plans with thresholds, B2B volume discounts)
- **Two-part tariffs:** fixed fee F (lump-sum) + per-unit price p (e.g. Costco: membership fee + retail prices; subscription + usage)
- **Bundling:** selling packages of products (e.g. Office 365, Netflix subscription, fast-food meal deals)
- **Versioning:** same base, versions at different prices (Word Standard vs Professional, airline service classes, hardcover/paperback editions)

Unifying logic

In all these forms, the monopolist *induces* consumers to reveal their WTP through their choices. The constraint is not information (as in the first degree), but each type's optimal behavior: each consumer must prefer the option intended *for him or her* (*incentive compatibility constraint*, IC).

Two-Part Tariff: Homogeneous Consumers

Setup. N identical consumers, each with individual demand $q(p)$. The monopolist offers a tariff (F, p) : entry fee F + unit price p .

Participation constraint (PC). The consumer buys only if $CS(p) \geq F$, where $CS(p) = \int_0^{q(p)} (P_D(q) - p) dq$ is the surplus at price p .

Monopolist's profit (over N consumers):

$$\pi(F, p) = N \cdot F + N \cdot (p - MC) \cdot q(p)$$

The monopolist maximizes by setting $F = CS(p)$ (binding PC: maximum extraction):

$$\pi(p) = N \cdot CS(p) + N \cdot (p - MC) \cdot q(p)$$

Key result

Differentiating with respect to p and using $dCS/dp = -q(p)$:

$$\frac{d\pi}{dp} = N \cdot [-q(p) + q(p) + (p - MC) q'(p)] = N(p - MC) q'(p) = 0$$

Since $q'(p) < 0$, the only solution is $p^* = MC$. One sets $F^* = CS(MC)$ and captures all the surplus.

Equivalent to first-degree discrimination.

Two-Part Tariff: Heterogeneous Consumers (1/2)

Setup (Oi 1971). Two types of consumers, in equal numbers. For simplicity $MC = 0$:

Type H (high WTP): $q_H = 20 - p$, Type L (low WTP): $q_L = 12 - p$

The monopolist must offer a *single* tariff (F, p) to both (it cannot discriminate by type, otherwise it would be third degree).

Trade-off. To serve both, $F \leq CS_L(p)$ (L's participation constraint, the more restrictive one). Hence:

$$F = CS_L(p) = \frac{1}{2}(12 - p)^2$$

Case 1: serve both types. Profit on a pair of consumers (1 H + 1 L):

$$\pi(p) = 2F + p \cdot (q_H + q_L) = (12 - p)^2 + p \cdot (32 - 2p) = 144 + 8p - p^2$$

$$\frac{d\pi}{dp} = 8 - 2p = 0 \implies p^* = 4, \quad F^* = \frac{1}{2}(12 - 4)^2 = 32$$

$$q_H^* = 16, \quad q_L^* = 8, \quad \pi_{\text{both}}^* = 144 + 32 - 16 = 160$$

Two-Part Tariff: Heterogeneous Consumers (2/2)

Case 2: exclude type L, serve only H. The monopolist applies the homogeneous-case formula to type H alone:
 $p = MC = 0$, $F = CS_H(0) = \frac{1}{2} \cdot 20 \cdot 20 = 200$.

Profit on the pair (with L excluded):

$$\pi_{\text{only H}}^* = 200 + 0 = 200$$

Comparison: $200 > 160$. It pays to exclude type L.

Counterintuitive insight (Oi, 1971)

When types are very heterogeneous, the monopolist prefers to *give up* the low type and extract all the surplus of the high types (which is larger). Serving both requires keeping F low, losing a lot on the high types.

When does it pay to serve both? The choice depends on the *calibration*:

- If types are *similar*, it pays to serve both (small loss on H)
- If types are *very different*, it pays to exclude L
- In practice: amusement parks, gyms, clubs offer *menus* of tariffs (e.g. family vs single) to approximate third-degree discrimination

Bundling: Selling Goods in Packages

Bundling

The monopolist sells two (or more) goods only *together*, at a bundle price p_{AB} . **Pure bundling**: only the bundle. **Mixed bundling**: bundle and individual goods offered simultaneously.

When is it profitable? (Adams & Yellen, *QJE* 1976) When consumers' valuations of the two goods are **negatively correlated**: bundling *homogenizes* total WTP, reducing dispersion and allowing more surplus to be extracted.

Numerical example. 2 goods A, B ; 2 consumers; $MC = 0$ for both. WTP:

	WTP for A	WTP for B	WTP for $A + B$
Consumer 1	10	4	14
Consumer 2	4	10	14

Separate pricing. For A : at $p_A = 10$ I sell only to 1 $\rightarrow$ revenue 10; at $p_A = 4$ I sell to both $\rightarrow$ revenue 8. $p_A = 10$ is better. Symmetric for B . **Total profit with separate pricing** = $10 + 10 = 20$.

Pure bundling. Both value $A + B$ at exactly 14. I set $p_{AB} = 14$: both buy. **Total profit** = $14 \times 2 = 28$.
 $\Rightarrow$ Bundling increases profit by 40% (+8).

Bundling: Why It Works

Statistical logic. Individual WTPs for A and B are *negatively correlated*: whoever values A more values B less, and vice versa. Bundling reduces the variance of aggregate WTP.

In our example: $\text{Var}(WTP_A) = \text{Var}(WTP_B) = 9$, but $\text{Var}(WTP_{A+B}) = 0$. The WTP for the bundle is *identical* for both: the monopolist can set $p_{AB} = 14$ and extract everything.

Mixed bundling. In more realistic cases (correlation not perfectly negative), it pays to offer the bundle and the individual goods *simultaneously* at different prices. Consumers self-select:

- Those who value both goods in a balanced way $\rightarrow$ bundle
- Those who value A highly and B little (or vice versa) $\rightarrow$ single good

Real-world examples

- **Microsoft Office:** Word+Excel+PowerPoint package at $\sim \$70/\text{year}$; individual products not sold separately (pure bundling)
- **Multiplex cinemas:** unlimited monthly pass vs. single ticket (mixed)
- **Fast-food meal deals:** burger+fries+drink at a price below the sum (mixed)
- **Pay-TV:** sports+movies channel package vs. individual channels à la carte

Welfare Table: Comparing Regimes

Qualitative ranking of the regimes in terms of CS , PS , TS , and DWL .

Regime	Q	CS	$PS (\pi)$	TS	DWL
Perfect competition	Q_{pc}	high	~ 0	maximum	0
Single-price monopoly	$Q_m < Q_{pc}$	reduced	positive	reduced	> 0
Third-degree discr.	$Q_m \leq Q_{III} < Q_{pc}$	reduced	$> \text{unif. mon.}$	ambiguous	ambiguous
Two-part tariff (homog.)	Q_{pc}	≈ 0	$\approx TS_{pc}$	maximum	0
First-degree discr. (perfect)	Q_{pc}	0	TS_{pc}	maximum	0
Mon. + rent-seeking ($\delta=1$)	Q_m	reduced	0 (diss.)	much reduced	triang. + rent

Reading the table

- **Efficiency:** first degree, homog. two-part tariff and PC are all efficient ($DWL = 0$). They differ in the *distribution* of surplus.
- **Equity:** PC maximizes CS , first degree wipes it out. The other forms lie in between.
- **Rent-seeking:** worst case, it destroys both the triangle and the rent.

Welfare Table: Numerical Example ($P = 100 - Q$, $MC = 20$)

Regime	Q	P	CS	PS	TS	DWL
Perfect competition	80	20	3,200	0	3,200	0
Single-price monopoly	40	60	800	1,600	2,400	800
First-degree discr.	80	$\in [20, 100]$	0	3,200	3,200	0
Two-part tariff, homogeneous	80	$p = 20, F = CS$	≈ 0	$\approx 3,200$	3,200	0
Mon. + rent-seeking ($\delta = 1$)	40	60	800	0	800	2,400

Key calculations: PC $\rightarrow P=MC=20$, $Q=80$, $CS=\frac{1}{2} \cdot 80 \cdot 80 = 3,200$. Unif. mon. $\rightarrow MR=100-2Q=20$, $Q=40$, $P=60$, $CS=\frac{1}{2} \cdot 40 \cdot 40 = 800$, $PS=40 \cdot 40=1,600$, $DWL=800$. First-degree discr. $\rightarrow Q=80$, $\pi = \int_0^{80} (80-Q) dQ = 3,200$. Mon. + rent-seeking ($\delta=1$) $\rightarrow \pi_m=1,600$ dissipated, $DWL_{\text{eff}} = 800 + 1,600 = 2,400$.

Crucial comparison

First-degree discrimination and the two-part tariff with homogeneous consumers *eliminate* the DWL but wipe out CS. Rent-seeking turns a moderate distortion ($DWL=800$) into severe waste ($DWL_{\text{eff}}=2,400$).

Exercise 1 — Third-Degree Discrimination

A monopolist serves two separate markets. Total cost is $C(Q_A + Q_B) = 8(Q_A + Q_B) + 100$.

- Market A (households): $P_A = 60 - Q_A$
- Market B (firms): $P_B = 40 - Q_B/2$

- Write MR_A , MR_B , and MC .
- Find the optimal quantities Q_A^* , Q_B^* and prices P_A^* , P_B^* .
- In which market is the price higher? Check consistency with the Lerner rule.
- Compute total profit.

Exercise 1 — Solution (a) and (b)

- (a)** Inverse demand A : $P_A = 60 - Q_A \Rightarrow TR_A = 60Q_A - Q_A^2 \Rightarrow MR_A = 60 - 2Q_A$.
Inverse demand B : $P_B = 40 - Q_B/2 \Rightarrow TR_B = 40Q_B - Q_B^2/2 \Rightarrow MR_B = 40 - Q_B$.
Marginal cost: $MC = 8$.

(b) $MR_A = MC$:

$$60 - 2Q_A = 8 \implies Q_A^* = 26, \quad P_A^* = 60 - 26 = 34$$

$MR_B = MC$:

$$40 - Q_B = 8 \implies Q_B^* = 32, \quad P_B^* = 40 - 16 = 24$$

Exercise 1 — Solution (c) and (d)

(c) $P_A^* = 34 > P_B^* = 24$: households pay more.

Check with the Lerner rule. We compute $|\varepsilon_D|$ at the optimum:

$$|\varepsilon_{D,A}| = \frac{dQ_A}{dP_A} \cdot \frac{P_A}{Q_A} = 1 \cdot \frac{34}{26} \approx 1.31$$

$$|\varepsilon_{D,B}| = 2 \cdot \frac{24}{32} = 1.5$$

Demand is less elastic in market A ($1.31 < 1.5$): the equilibrium price is higher in A ✓.

Lerner index: $L_A = (34 - 8)/34 \approx 0.76$ and $L_B = (24 - 8)/24 \approx 0.67$. Consistently, $L_A > L_B$.

(d) Total profit:

$$\pi_A = 26 \cdot 26 = 676, \quad \pi_B = 16 \cdot 32 = 512, \quad \pi_{\text{tot}} = 676 + 512 - 100 = 1,088$$

Exercise 2 — Two-Part Tariff (Homogeneous)

An amusement park has N *identical* visitors. Each has individual demand $q = 20 - p$ for rides, where p is the price per ride. The marginal cost of a ride is $MC = 4$.

Management introduces a *two-part tariff*: park admission at price F (lump-sum) + price p for each ride.

- (a) Which unit price p^* maximizes profit (and why)?
- (b) Compute each consumer's CS at price p^* . How much can the park charge as admission F^* without driving visitors away?
- (c) Profit per visitor with the two-part tariff vs. single monopoly price: compare.

Exercise 2 — Solution

(a) With identical consumers, it pays to set $p^* = MC = 4$: every unit with $WTP > MC$ is traded, $DWL = 0$.
At $p^* = 4$: $q^* = 20 - 4 = 16$ rides per visitor.

(b) The individual consumer's CS at price $p^* = 4$:

$$CS = \frac{1}{2} \cdot q^* \cdot (p_{\max} - p^*) = \frac{1}{2} \cdot 16 \cdot (20 - 4) = 128$$

The park can charge $F^* = 128$ as admission: the consumer is indifferent between entering (gets $CS = 128$, pays $F = 128$, net surplus 0) and not entering. In practice F^* is set slightly below 128 to ensure participation.

(c) **Profit per visitor with the two-part tariff:** $\pi = F + (p^* - MC) \cdot q^* = 128 + 0 = 128$.

Comparison with the single monopoly price. $MR = 20 - 2q = MC = 4 \Rightarrow q_m^* = 8, p_m^* = 12$,

$$\pi_m = (12 - 4) \cdot 8 = 64.$$

The two-part tariff *doubles* profit and *eliminates* the DWL. It captures all the CS without distorting quantity: Pareto efficiency *and* maximum profit.

Exercise 3 — First-Degree Discrimination

A monopolist faces market demand $P = 80 - Q$ and cost function $C(Q) = 8Q$ (so $MC = 8$, $FC = 0$).

- (a) Compute the monopoly price, quantity, and profit with a *single price*. Compute CS_m and DWL_m .
- (b) Now suppose the monopolist can practice *first-degree discrimination*. Find the quantity produced Q^{disc} and the profit π^{disc} . (Hint: π^{disc} is the *triangular area* between the demand curve and the MC line, from 0 to Q^{disc} .)
- (c) Verify that $\pi^{disc} - \pi_m = CS_m + DWL_m$ from case (a). Interpret.

Exercise 3 — Solution (a)

(a) **Single monopoly price.** $MR = 80 - 2Q = MC = 8$:

$$Q_m^* = \frac{80 - 8}{2} = 36, \quad P_m^* = 80 - 36 = 44$$

$$\pi_m = (P_m^* - MC) \cdot Q_m^* = (44 - 8) \cdot 36 = 36 \cdot 36 = 1,296$$

Consumer surplus.

$$CS_m = \frac{1}{2} \cdot Q_m^* \cdot (P_{\max} - P_m^*) = \frac{1}{2} \cdot 36 \cdot (80 - 44) = \frac{1}{2} \cdot 36 \cdot 36 = 648$$

DWL. Under PC $P = MC = 8 \Rightarrow Q_{pc} = 72$. Hence:

$$DWL_m = \frac{1}{2} (Q_{pc} - Q_m^*) (P_m^* - MC) = \frac{1}{2} \cdot 36 \cdot 36 = 648$$

(In this symmetric example, $CS_m = DWL_m$. It is a special case of linear demand with P_m^* halfway between $P_{\max}$ and MC .)

Exercise 3 — Solution (b) and (c)

(b) First-degree discrimination. The monopolist sells each unit at the price on the demand curve as long as $P_D(Q) \geq MC$, hence up to $Q^{disc} = Q_{pc} = 72$.

Profit is the **area of the triangle** between the demand curve and the MC line, from quantity 0 to Q^{disc} :

$$\pi^{disc} = \frac{1}{2} \cdot \underbrace{Q^{disc}}_{\text{base}} \cdot \underbrace{(P_{\max} - MC)}_{\text{height}} = \frac{1}{2} \cdot 72 \cdot (80 - 8) = \frac{1}{2} \cdot 72 \cdot 72 = 2,592$$

(c) Check. $\pi^{disc} - \pi_m = 2,592 - 1,296 = 1,296$, and $CS_m + DWL_m = 648 + 648 = 1,296 \checkmark$.

Interpretation

First-degree discrimination allows the monopolist to capture *exactly*:

- the consumer surplus of the uniform model ($CS_m = 648$), transferring it to itself;
- the area of the trades that uniform monopoly did *not* carry out ($DWL_m = 648$), which are now carried out.

The perfectly discriminating monopolist leaves no surplus to consumers *and* wastes no mutually beneficial trades.

General Summary (1/2)

Social costs of monopoly revisited

- The rent π_m may not be a mere transfer: if firms compete to obtain it, it is dissipated (rent-seeking)
- Tullock model: $X^* = \frac{n-1}{n} V$, dissipation $\rightarrow V$ for large n
- $DWL_{\text{eff}} = \text{Harberger triangle} + \delta \cdot \pi_m$: the social cost can be much larger than the triangle alone

Lerner index

- $L = (P - MC)/P \in [0, 1]$: measures the relative markup
- At the monopolist's optimum: $L = 1/|\varepsilon_D|$
- The more inelastic demand is, the more the monopolist raises the price

First-degree (perfect) discrimination

- A different price for each unit of each consumer (= to the WTP)
- The monopolist captures all the CS; $Q = Q_{pc}$, so $DWL = 0$
- Maximum efficiency, maximum inequity

General Summary (2/2)

Third-degree discrimination (groups)

- Different prices for identifiable groups (students, business, countries, ...)
- Optimum: $MR_i = MC$ for each market i
- Higher price where $|\varepsilon_D|$ is lower (Lerner formula market by market)

Second-degree discrimination (nonlinear prices)

- Quantity discounts, two-part tariffs, bundling, versioning
- Two-part tariff, homogeneous: $p^* = MC$, $F^* = CS$ (equivalent to first degree)
- Two-part tariff, heterogeneous: trade-off between excluding low types vs. serving both (Oi 1971)
- Bundling: effective when WTPs are negatively correlated (Adams-Yellen 1976)

Necessary conditions for discrimination

- Market power (price-maker)
- Information about consumers' types/WTP (for first and third degree)
- No arbitrage between groups (resale impossible)

Skills Acquired

- ✓ Distinguish monopoly as a transfer from monopoly with rent-seeking
- ✓ Solve the Tullock model and compute the fraction of rent dissipated
- ✓ Compute the Lerner index $L = (P - MC)/P$ and link it to the elasticity of demand
- ✓ Distinguish the three degrees of discrimination (Pigou: first, second, third) and their conditions
- ✓ Explain why first-degree discrimination eliminates the DWL but wipes out the CS
- ✓ Solve a third-degree discrimination problem using $MR_1 = MR_2 = MC$
- ✓ Construct an optimal two-part tariff (homogeneous and heterogeneous consumers)
- ✓ Recognize when bundling is profitable (negative correlation of WTPs)
- ✓ Compare welfare under the different pricing regimes (PC, monopoly, first/second/third degree, rent-seeking)
- ✓ Recognize real-world applications (cinemas, flights, supermarkets, software, lobbying, patents)

Optional Homework Exercises

1. Monopolist with $P = 100 - 2Q$ and $MC = 20$. Compute Q_m^* , P_m^* , the Lerner index, and check it against $1/|\varepsilon_D|$.
2. *Third-degree discrimination*. A university sets different tuition fees for Italian and foreign students. Demand of Italian students: $P_I = 6,000 - 2Q_I$. Demand of foreign students: $P_S = 10,000 - 4Q_S$. Marginal cost per student: $MC = 1,000$. Find the optimal fees and enrollments for the two groups.
3. *First-degree discrimination*. Market demand $Q = 60 - P$, $MC = 10$. (a) Compute π_m with a single price. (b) Compute π^{disc} with first-degree discrimination. (c) Verify that $\pi^{disc} - \pi_m = CS_m + DWL_m$.
4. *Two-part tariff with heterogeneity*. Type A: $q_A = 12 - p$. Type B: $q_B = 8 - p$. $MC = 0$. Equal numbers of each type. Determine the optimal tariff (F, p) if the monopolist wants to serve both types, and compare with the strategy of serving only type A.
5. *Rent-seeking*. Three firms compete à la Tullock for a monopoly rent $V = 1,000$. Compute the individual investment x^* , aggregate spending X^* , and the fraction of rent dissipated. Discuss how the result would change with $n = 10$ firms.
6. True or false? Explain: “with third-degree discrimination between two markets, the higher price applies to the market where demand is flatter”.